If you isolate **only the purely linear predictors** in SARTA, the entire matrix collapses down to just **four baseline variables**. [1]

In a strictly linear sense, SARTA does not use any powers (like squares or cubes), interaction products (like multiplying two variables together), or square roots. The strictly linear predictor set consists entirely of: [2]

- **$\sec(\theta)$**: The baseline slant path length (the geometric viewing zenith angle).
- **Δ T**: The absolute kinetic temperature deviation from the reference profile layer ($T_{\text{actual}} - T_{\text{ref}}$).
- **W_r**: The linear water vapor abundance ratio ($Q_{\text{actual}} / Q_{\text{ref}}$).
- **O_r**: The linear ozone mass abundance ratio ($V_{\text{actual}} / V_{\text{ref}}$). [3]

Why SARTA Can't Rely *Only* on Linear Predictors

If SARTA only used these four linear terms, it would be a purely **multilinear regression model**. While it would compute instantly, it would suffer from large brightness temperature errors because the underlying physics of radiative transfer are highly non-linear: [4]

- **Missing Saturation Effects**: Without the non-linear square-root term ($\sqrt{W_r}$), it could not accurately model gas lines that flatten out and saturate. [5]
- **Missing Collision Effects**: Without the non-linear squared term (W_r^2), it would fail to track water vapor continuum collisions, which scale exponentially with humidity.

To capture those phenomena while keeping the code fast, SARTA creates a compromise: it takes these baseline linear inputs and expands them into a static suite of non-linear algebraic terms. The regression itself remains linear (solving Y = AX), but the *predictors* (X) are a mixture of linear base variables and non-linear polynomial expansions. [4, 5, 6]

Would you like to look at the **exact regression matrix equation** showing how these terms are combined with SARTA's static coefficients, or examine how its **tangent-linear model code** isolates these linear sensitivities? [7, 8]

[1] https://www.ecmwf.int
[2] https://www.youtube.com
[3] https://repository.library.noaa.gov
[4] https://www.spiedigitallibrary.org
[5] https://www.sciencedirect.com
[6] https://www.sciencedirect.com
[7] https://journals.ametsoc.org
[8] https://www.sciencedirect.com